ECE 3040 - Practice Test

1. **(7 points)** Consider a face-centered cubic (fcc) lattice:

   (a) (2 points) Draw a 3D primitive unit cell structure

   (b) (2 points) Sketch the placement of atoms on a (100) plane. Express distances between atoms on the plane in terms of lattice constant (a).

   (c) (1 point) How many atoms are there per primitive unit cell?

   (d) (1 point) How many nearest neighbor atoms are there for each atom?

   (e) (1 point) Assume a lattice constant of 1nm. Determine the number of atoms/cm^3

2. **(5 points)** A silicon wafer is doped with $10^{15}\ cm^{-3}$ boron atoms.

   (a) (1 point) Is the wafer n- or p-type?

(b) (2 points) Calculate the electron and hole densities at 0 K.

(c) (2 points) Calculate the electron and hole densities at room temperature.

3. **(4 points)** What is the dominant current mechanism in

(a) (2 points) forward biased p-n junction

(b) (2 points) reverse biased p-n junction

4. **(13 points)** Consider a p-n step junction under equilibrium with **p-side more heavily doped**. Label all necessary parameters on your sketches.

(c) (3 points) Sketch charge density distribution as a function of position

(d) (3 points) Sketch the electric field as a function of position

(e) (3 points) Sketch the electrostatic potential as a function of position

(f) (4 points) Sketch the energy band diagram labeling E_C, E_V, E_i and E_F on both sides of the junction

5. **(6 points)** Assume that a p-type and an n-type silicon semiconductor are nondegenerately doped with the same doping concentration. Between two samples, the p-type has higher resistivity than the n-type at room temperature.

(a) (3 points) How does electron and hole mobilities compare? Explain.

(b) (3 points) How does electron and hole diffusion coefficients compare? Explain.

6. **(5 points)** Why are direct bandgap materials more efficient in absorbing light compared to indirect bandgap materials? Explain

7. **(4 points)** Which capacitance dominates the small signal behavior of a reverse biased p-n junction and what is the origin of this capacitance?

8. **(6 points)** Consider a p-n junction

(a) (3 points) What is the physical mechanism leading to an avalanche breakdown in a p-n junction? Explain the mechanism.

(b) (3 points) How does breakdown voltage in a reverse-biased p-n junction change with doping concentration of the lightly doped side? Explain why?

Problem (50 points): A silicon step junction at T = 300 K has doping concentrations of $N_A = 5x10^{16}\ cm^{-3}$ and $N_D = 5x10^{15}\ cm^{-3}$. The cross-sectional area of the junction is 100 μm^2.
Show all steps in your solution

(a) (5 points) Calculate the Fermi energy on each side of the junction (in the quasi-neutral regimes) with respect to the intrinsic Fermi level, i.e. E_F - E_i.

(b) (5 points) Calculate the resistivity of both n-type and p-type quasi-neutral regions.

(c) (5 points) Calculate the built-in potential V_{bi}.

(d) (5 points) Calculate the width of the depletion region under zero bias. What percentage of the depletion region is in the n-side of the junction?

(e) (5 points) By what factor does the depletion region width increase if you reverse bias the junction by at V_A = -25 V?

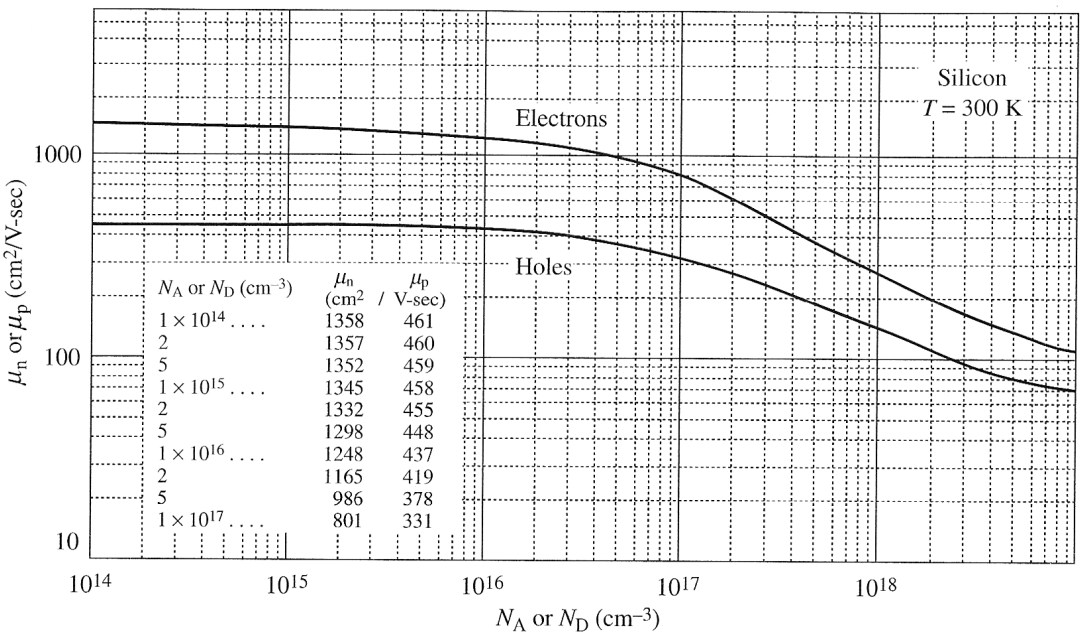
(f) (5 points) Calculate junction capacitance of p-n junction reverse biased at V_A = -25 V.

(g) (10 points) Now assume that the p-n junction is forward biased with V_A = 0.5 V. Sketch, in log scale, the steady-state majority and minority carrier concentrations on both sides of the depletion region as a function of the position. Indicate the equilibrium level of minority carrier concentrations in your sketch.

(h) (10 points) Calculate the current density flowing through the p-n junction in case of a forward-bias voltage of V_A = 0.5 V. Assume a minority carrier lifetime for holes and electrons of $\tau_n = \tau_p$ = 1 μs.

You might need:

$q = 1.6x10^{-19}$ C, $K_s = 11.8$, $\varepsilon_0 = 8.85x10^{-14}$ F/cm, $kT = 0.026$ eV at T = 300K
$E_g = 1.12$ eV, $n_i = 10^{10} cm^{-3}$ at T = 300K, use carrier mobilities from the graph below.



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